

THE RELATIONSHIP MODEL

Topic 6

LOs

- By the end of this topic you should be able to:
 - Define the concepts that exist in a relationship model;
 - Differentiate between the existing key and integrity; and
 - Construct a relationship scheme.

Introduction

- We will shift our discussion from the process of conceptual design (ERD) to logical design.
- The logical design is the process of interpreting the conceptual data model to the logical data model
- The logical data model is used in the application of a database management system
- It is more flexible and easier to understand compared to the hierarchy data and network data models

THE DATA RELATIONSHIP MODEL

- The data relationship model was introduced in 1970 by Codd.
- The model was constructed based on **mathematical theory**.
- It represents data in the form of **tables**.
- The following sub-topic will discuss this and you will also be given explanations regarding the characteristics of a good relationship

The Structure of Relationship Data

- Data (known in the ER Model as entity) is represented as relationships in a relationship model.
- Relationship is a two dimensional table (made up of rows and columns) with names.
- Columns in a table represent the relationship attribute and rows contain record or attribute values known as *tuple*.
- There are several alternative terms for relationship, tuple and attribute.

The Structure of Relationship Data

- **Table 6.1:** Alternative Terms for Relationship, Tuple and Attribute

Term	Alternative Term	Alternative Term
Relationship	Table	File
Tuple	Row	Record
Attribute	Column	Field

The Structure of Relationship Data

- The relationship structure is portrayed in two simple and common format. This format is composed of **brief text statement** and **graphic representation**
- **Brief text statement:** Relationship name and its list of attributes is placed in brackets. The format is:
 - RELATIONSHIP NAME (Attribute 1, Attribute 2,, Attribute N)
 - An example is;
 - STUDENT (Metric_No., Name, Faculty, Address)

The Structure of Relationship Data

- **Graphic Representation:** Relationship name is placed on top of rectangular boxes containing the name of the relationship attribute. The format is:
- RELATIONSHIP NAME

Attribute 1	Attribute 2	Attribute 3	Attribute ...	Attribute N
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Example
STUDENT

Matric_No	Name	Faculty	Address	PhoneNo
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Relationship Key

- Similar to the E-R model concept, the key is used to identify each tuple (entity occurrence in the E-R model) uniquely.
- Each relationship must have a key known as the **primary key**.
- Always, the key in the E-R model will become the key in the relationship model.
- In certain cases such as composite entity and weak entity, different keys will be fixed.
- This exception will be explained in the next unit. The primary key can consist of one (or more) attribute. The primary key which consists of more than one attribute is known as **composite key**. The primary key is underlined to differentiate it from other attributes which are not keys.

Relationship Key

- How do we represent the connection between two relationships (or tables)? We must use one more key for this purpose and this key is known as **foreign key**.
- A **foreign key** is an attribute (or several attributes) in a relationship that becomes a primary key in other relationships.

Relationship Key

- For example, there are two relationships; STUDENT and COURSE.
 - STUDENT (Metric_No., Name, Faculty_Code, Course_Code)
 - COURSE (Course_Code, Course_Name, Credit_Hour)
- The Course_Code attribute is the foreign key in a STUDENT relationship.
- It enables users to connect the STUDENT to the COURSE he is taking.

Relationship Characteristics

1. Each relationship (or table) has a unique name.
2. Each data entry to the cell (the crossing of line and column) is **atomic** because the relationship does not have multiple value attributes.
3. Each row is unique. There are no two same rows in a particular relationship.
4. Each attribute (or column) in a relationship has a unique name.
5. The arrangement of columns is not important (or does not have any significant meaning). The arrangement of columns can be changed without changing the meaning of the relationship.
6. The arrangement of rows is not important (or does not possess any meaning). The arrangement of rows can also be changed without altering the meaning of the relationship.

Relationship Characteristics

- Observe the characteristic of the second relationship.
- If we define multiple value attributes in E-R model, what must we do during the translation into the relationship model?
- Because the relationship model does not have multiple value attributes, we need to delete it from the table.
- Actually, we do not 'delete' it according to the meaning of the word 'delete'. It still exists.
- We merely change its shape so that it is suitable to the definition of relationship.

Relationship Characteristics

- If you are still unclear as to what is being presented, do not worry.
- Let us examine the explanation which follows
- Assume we have a STUDENT relationship with attributes
 - Metric_No., Student_Name, Department and Major (Specialisation) as in Figure 6.3a.
- Because the student can have more than one Major, the Major attribute becomes a multiple value attribute.

STUDENT

Metric No	Name	Address	Major
A1000	Maisarah	College 1	Computer Science
A1001	Maria	College 1	Computer Science , Mathematics
A1002	Mariam	Bangi	Usuluddin
AlO03	Muaz	Kajang	Account Management
AlO04	Muhammad	Serdang	Chemistry Physics

Figure 6.3a: Figure with multiple values

2. Each data entry to the cell (the crossing of line and column) is **atomic** because the relationship does not have multiple value attributes.

STUDENT₂

Metric No	Name	Address	Major
AlOo0	Maisarah	College 1	Computer Science
AlOo1	Maria	College 1	Computer Science
AlOo1	Maria	College 1	Mathematics
AlOo2	Mariam	Bangi	Usuluddin
AlOo3	Muaz	Kajang	Management
AlOo3	Muaz	Kajang	Accounting
AlOo4	Muhammad	Serdang	Chemistry
AlOo4	Muhammad	Serdang	Physics

Figure 6.3b: Student₂ relationships

- To delete a multiple value attribute, break up the multiple value attribute. (example Major attribute) into individual attributes as shown in Figure 6.4b.
- As such, the table in Figure 6.4b only contain solitary attributes and it has fulfilled the atomic character of relationship.
- The table name has been changed to STUDENT₂ to distinguish it from the original STUDENT table.
- The new table consists of weaknesses that will be rectified in the normalisation topic which you will learn in the next unit.

INTEGRITY CONSTRAINT

- The relationship data model has several types of constraints to **ensure accuracy** and **data integrity** in a database.
- Among them are **domain constraint**, **entity integrity**, **reference integrity** and **organisational constraint**.

Domain Constraint

- A domain normally contains the following elements:
 - Domain Name
 - Description
 - Data type
 - Data Size (length)

Domain Constraint

Attribute	Domain Name	Explanation	Domain: Data Type	Size
Metric_No.	Metric_No.	Student metric number	Character	5
Student_Name	Student_Name	Student name	Character	20
Student_Address	Student_Address	Student address	Character	30
Course_Code	Course_Code	Code of course offered	Character	5
Course Name	Course Name	Name of course offered	Character	20
Scholarship_Code	Scholarship_Code	Student scholarship code	Numeric	5
Scholarship Name	Scholarship Name	Scholarship name	Character	2
Date_Begin	Date_Begin	Date scholarship offered	Date Mm/dd/yy	
Date End	Date End	Date scholarship ends	Date Mm/dd/yy	
Total Scholarship	Total Scholarship	Total scholarship	Float	10

Table 6.2: Example of Domain Definition

Entity Integrity

- The purpose of entity integrity is to ensure that each relationship possesses the primary key and data value that is legal.
- What is the legal data value for the primary key? Do you still remember the attribute conditions to be chosen as primary key?
- If a particular attribute is to be made a key, it **must not** have the null value.
- You also know that the **null value** is placed on attributes if the attribute value is unknown at the time it is to be filled or if its attribute value bears no relation to the situation or the entity itself. **The null value is not the same as the number zero (0) or blank character.**

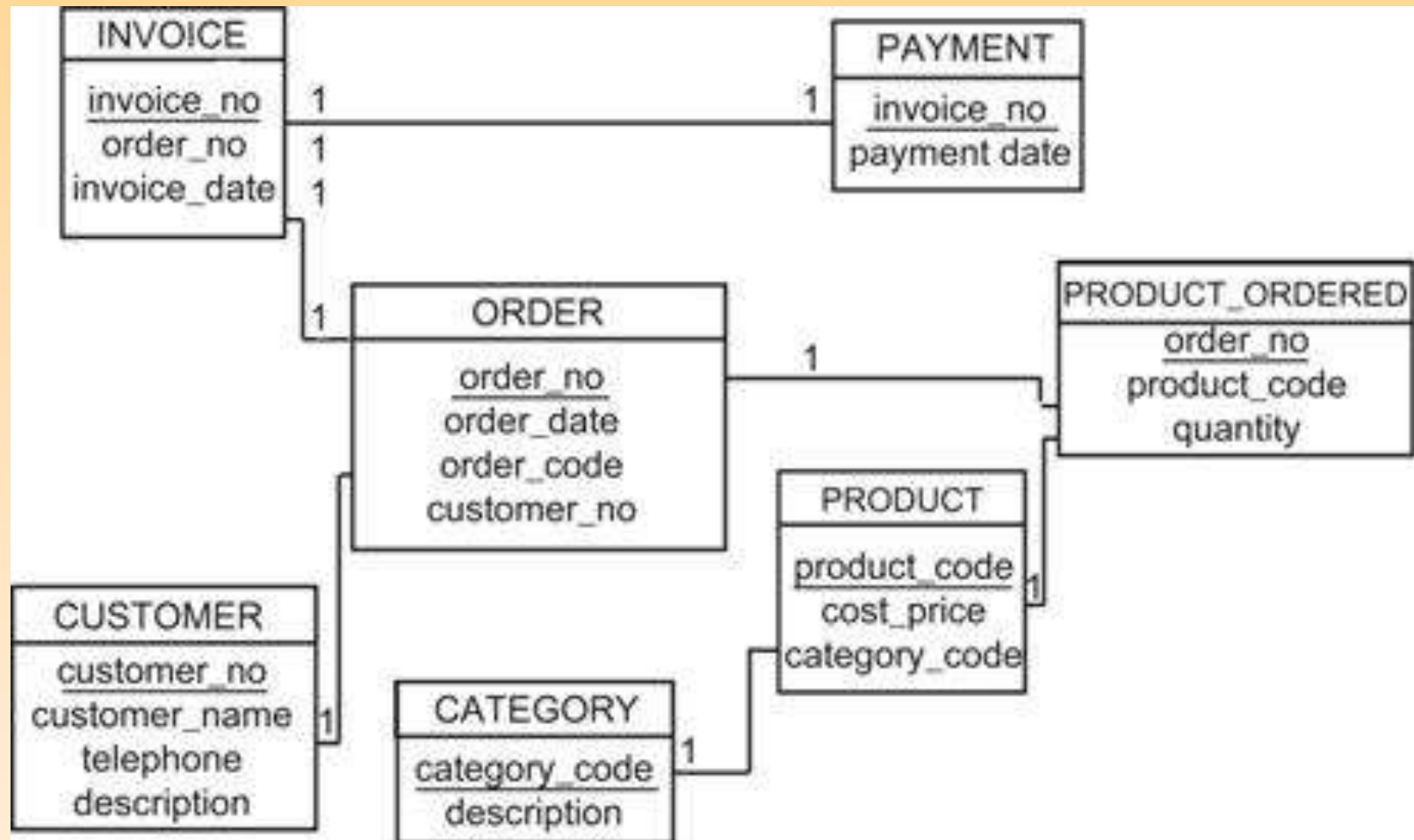
Entity Integrity

- The entity integrity makes it **compulsory for you to fill each column of key attribute** (with unique values).
- As such, the entity integrity states that the primary key must not contain the null value.
- Why? Because, if the null value is permitted, there may be several key values with the null value.
- As such the key value is no longer unique.

Reference Integrity

- The relationship between tables in a relationship model is represented by a foreign key.
- You have been given explanations regarding foreign key before this.
- Logically, two tables are related through two attributes having the same domain.
- Figure 6.5 shows this table relationship. Notice that each foreign key matches the primary key

Reference Integrity



Reference Integrity

- The purpose of reference integrity is to ensure that data rows are consistent between related tables.
- The reference integrity states that if a foreign key exists in a relationship, then:
 - 1. The value of each foreign key must match the value of each primary key in the relationship it represents
 - OR
 - 2. The value of the foreign key must be null.

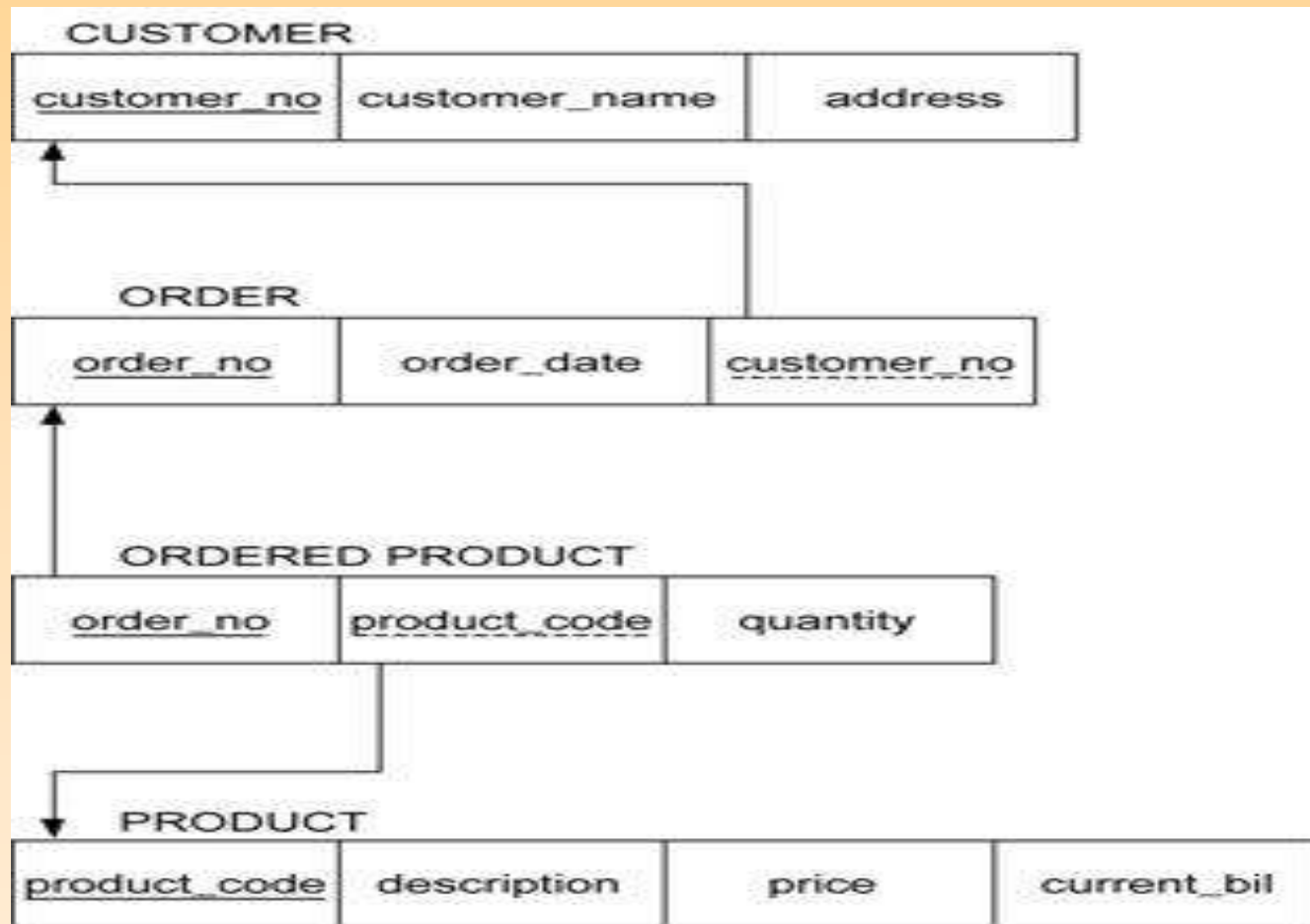
Reference Integrity

- For example, Department_Code is a foreign key in the relationship WORKER.
- Department_Code is the primary key in its original relationship i.e. Department.
- We cannot, for example, record worker particulars by using department code o6 if the department code is not yet in existence in the Department relationship.
- However, we can fill the Department_Code space with the null value for specific cases, for example, if new workers are appointed and they have not yet been assigned to specific departments.

Reference Integrity

- We can examine where the reference integrity need to be checked by representing graphically the relationship between tables.
- List the relationship scheme and place arrows in between the foreign key and the primary key according to the related tables.
- Please observe closely Figure 6.6 below to obtain a clearer picture of the representation of this table.

Reference Integrity



Organisation Constraints

- Organisation constraints are statements in the form of rules which show the characteristics or constraints of the business.
- For example: "If a pre-degree student intends to fulfil the conditions of becoming a degree holder he must accumulate 120 credit hours" or " If a customer buys more than RM 100 worth of goods, he will be given a discount of 20%."

Organisation Constraints

- How can a database assist in implementing this organisational constraint? This can be done through the entity domain and attribute by making use of data manipulation facilities and other aspects of DBMS.
- For example, when a clerk processes student status at the end of the semester, the database system used by him/her has already defined student status according his/her position.

Organisation Constraints

- For example, the SQL statement has already defined that if the CGPA is less than 2.00, the status is fail; if the CGPA is more than 2.00, the status is pass; and if the CPGA is more than 2.00 and credit hours more or the same as 2.00 the status is completed studies.
- As such, processed student data will contain the correct status according to the fixed organisational constraint.